COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 1 : Session – 2 : SLO – 1 | Digital systems and number systems
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MCQs

• 1. c — Combinational
• 2. c — Discrete physical quantities
• 3. d — Hexadecimal
• 4. b — Synchronous
• 5. c/d — 4 bits (rounded up from log2 10)

Two-Mark Questions

• 1. Synchronous sequential systems update their state only at fixed clock intervals (clock-driven, easier to design and debug); asynchronous systems change state whenever inputs change (no clock, faster but prone to race conditions and harder to analyze).
• 2. A binary signal is a signal that can take only one of two discrete values (0 or 1). Physical examples: voltage levels 0 V / +5 V in a circuit; switch OFF/ON; magnetic polarity on a storage disk.

Scenario-Based Long Answer

• Suitable system: a synchronous SEQUENTIAL digital system (a counter built from flip-flops).
• State: the current count is stored in flip-flops (internal state with memory). Input: count pulses from the wheel sensor and a reset line. Output: the state drives the odometer display.
• Synchronous is justified because all flip-flops update together on a common clock edge, so the count changes cleanly and predictably with no race conditions; an asynchronous ripple counter could show transient wrong values. A synchronous counter with a clear/reset input meets the precise-tracking and reset requirements.

Match the Following

• Binary — b) Base 2
• Hexadecimal — a) Base 16
• Byte — c) 8 bits
• Synchronous System — d) Updates at fixed intervals
• Asynchronous System — e) Updates at any time

Fill in the Blanks

• 1. combinational 2. octal 3. nibble 4. 0 and 1 5. Digital

Scenario-Based Small Activity

• A basic calculator uses BOTH: combinational logic (adders/ALU that compute results from current inputs) and sequential logic (registers/memory that hold operands and running totals).
• Components: (i) Inputs — keypad; (ii) Outputs — display; (iii) Memory/state — registers storing the first operand and intermediate result.
• Classification logic: the arithmetic unit alone is combinational (output depends only on present inputs), but storing numbers between key presses requires state, making the overall calculator a sequential system.

True or False

• 1. FALSE — A nibble is 4 bits. 2. FALSE — A digital signal takes only discrete values (0 or 1).
• 3. TRUE 4. FALSE — Sequential systems are usually synchronous. 5. TRUE